

Galgotias College of Engineering & Technology

Affiliated to Dr A.P.J AKTU, Lucknow

Department of Computer Science & Engineering



Question Bank

Design and analysis of algorithm

(BCS- 503)

(Odd Semester, 2024-25)

Name	: Mr. Manoj Kumar Dixit
Subject Name	: Design and Analysis of algorithm
Subject Code	: BCS-503
Semester	: B.Tech (CSE,CSE AI, CSE AIML, CSE DS, CSE CSD, AIML)

Department of Computer Science and Engineering

Faculty Name: ...Mr. Manoj Kumar Dixit.....Designation: ...AP III.....

Subject Name / Subject Code: DAA (BCS-503) Section: CSE

Checklist for Course File (Theory) # 2023 – 2024 (Even Semester)

S. No.	Description
1.	Cover Page
2.	Vision and Mission of the Institute
3.	Vision and Mission of the Department, PEOs, POs, PSOs
4.	Course Description with: i) CO, CO-PO/PSO Mapping, ii) Syllabus iii) Course Plan iv) Evaluation Scheme
5	Unit-wise question Bank
6.	Resource used



GALGOTIAS COLLEGE OF ENGINEERING AND TECHNOLOGY

Knowledge Park-II, Greater Noida, U.P.

INSTITUTE VISION AND MISSION

Vision of the Institute

To be a leading educational institution recognized for excellence in engineering education & research producing globally competent and socially responsible technocrats.

Mission of the Institute

- To provide state-of-the-art infrastructural facilities that support achieving academic excellence.
- To provide a work environment that is conducive for professional growth of faculty & staff.
- To collaborate with industry for achieving excellence in research, consultancy and entrepreneurship development.

DEPARTMENT VISION, MISSION & PEOS

Department Vision

To be center of excellence in providing contemporary education and research in the field of Computer Science and Engineering.

Department Mission

- **M1:** To create conducive learning environment through effective teaching and learning process promoting quality education and research
- **M2:** To promote research, consultancy and entrepreneurship development through industry - institute collaborations
- **M3:** To make the students corporate ready with the right spirit and interpersonal skills

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

- **PEO1:** Excel in their career by acquiring knowledge in the field of Computer Science & Engineering with the usage of modern tools and emerging technologies.
- **PEO2:** Demonstrate and analyze real life problems of the society and produce innovative solutions.
- **PEO3:** Exhibit amicable communication skills and team work in the industry and research organization through professional development and lifelong learning.



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PROGRAM OUTCOMES (POs)

Program Outcomes	Statement
PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex computer engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyse complex computer engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex computer engineering problems and design system components or processes that meet the specific needs with appropriate considerations for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent relevant to the professional engineering practices
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norm of the engineering practices
PO9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communications: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

Program Specific Outcomes (PSO)	Statement
	Engineering graduates will be able to:
PSO1	Design computer-based systems using theoretical computer science that demonstrate the comprehension of the trade-offs involve in the design choice.
PSO2	Design, develop and test software for different applications with real time constraints.



GALGOTIAS COLLEGE OF ENGINEERING AND TECHNOLOGY
Knowledge Park-II, Greater Noida, U.P.
Department of CSE & Allied Specialized Branches
Even Semester, 2023-24

FACULTY AND COURSE DETAIL

Faculty : Mr. Manoj Kumar Dixit

L T P C : 3 0 0 3

Course Type : CORE

Semester Offered : EVEN

Academic Year : 2023-24

Time : Mon: 11:10- 12:00 Noon

Tue: 09:30-10:20 AM

Wed: 10:20-11.10 AM

Thurs: 3:20-4:10 PM,

Class Room : C-210

Course Description:

Algorithm design and analysis provide the theoretical backbone of computer science and are a must in the daily work of the successful programmer. This course aims to provide a solid background in the design and analysis of the major classes of algorithms. At the end of the course, students will be able to develop their versions for a given computational task and to compare and contrast their performance.



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Department of CSE & Allied Specialized Branches

Even Semester, 2023-24

COURSE OUTCOMES

CO-POs AND CO-PSOs MAPPING MATRICES

Course Outcomes	Program Outcomes (PO)												Program Specific Outcomes (PSO)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
BCS-503.1	2	2	2	2	-	-	-	-	-	-	-	2	2	2
BCS-503.2	3	3	2	2	-	-	-	-	-	-	-	3	3	3
BCS-503.3	3	3	2	2	-	-	-	-	-	-	-	3	3	3
BCS-503.4	3	3	2	2	-	-	-	-	-	-	-	3	3	3
BCS-503.5	3	3	2	2	2	-	-	-	-	-	-	3	3	3
Average	2.8	2.8	2	2	2	-	-	-	-	-	-	2.8	2.8	2.8



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Department of CSE & Allied Specialized Branches

Even Semester, 2023-24

SYLLABUS

BCS503 DESIGN AND ANALYSIS OF ALGORITHM		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to understand		
CO 1	Design new algorithms, prove them correct, and analyze their asymptotic and absolute runtime and memory demands.	K ₄ , K ₆
CO 2	Find an algorithm to solve the problem (create) and prove that the algorithm solves the problem correctly (validate).	K ₅ , K ₆
CO 3	Understand the mathematical criterion for deciding whether an algorithm is efficient, and know many practically important problems that do not admit any efficient algorithms.	K ₂ , K ₃
CO 4	Apply classical sorting, searching, optimization and graph algorithms.	K ₂ , K ₄
CO 5	Understand basic techniques for designing algorithms, including the techniques of recursion, divide-and-conquer, and greedy.	K ₂ , K ₃
DETAILED SYLLABUS		3-1-0
Unit	Topic	Proposed Lecture
I	Introduction: Algorithms, Analyzing Algorithms, Complexity of Algorithms, Growth of Functions, Performance Measurements, Sorting and Order Statistics - Shell Sort, Quick Sort, Merge Sort, Heap Sort, Comparison of Sorting Algorithms, Sorting in Linear Time.	08
II	Advanced Data Structures: Red-Black Trees, B – Trees, Binomial Heaps, Fibonacci Heaps, Tries, Skip List	08
III	Divide and Conquer with Examples Such as Sorting, Matrix Multiplication, Convex Hull and Searching. Greedy Methods with Examples Such as Optimal Reliability Allocation, Knapsack, Minimum Spanning Trees – Prim's and Kruskal's Algorithms, Single Source Shortest Paths - Dijkstra's and Bellman Ford Algorithms.	08
IV	Dynamic Programming with Examples Such as Knapsack, All Pair Shortest Paths – Warshall's and Floyd's Algorithms, Resource Allocation Problem, Backtracking, Branch and Bound with Examples Such as Travelling Salesman Problem, Graph Coloring, n-Queen Problem, Hamiltonian Cycles and Sum of Subsets.	08
V	Selected Topics: Algebraic Computation, Fast Fourier Transform, String Matching, Theory of NP Completeness, Approximation Algorithms and Randomized Algorithms	08
Text books: 1. Thomas H. Cormen, Charles E. Leiserson and Ronald L. Rivest, "Introduction to Algorithms", Prentice Hall of India. 2. E. Horowitz & S Sahni, "Fundamentals of Computer Algorithms", 3. Aho, Hopcraft, Ullman, "The Design and Analysis of Computer Algorithms" Pearson Education, 2008. 4. LEE "Design & Analysis of Algorithms (POD)", McGraw Hill 5. Richard E. Neapolitan "Foundations of Algorithms" Jones & Bartlett Learning 6. Jon Kleinberg and Éva Tardos, Algorithm Design, Pearson, 2005. 7. Michael T Goodrich and Roberto Tamassia, Algorithm Design: Foundations, Analysis, and Internet Examples, Second Edition, Wiley, 2006. 8. Harry R. Lewis and Larry Denenberg, Data Structures and Their Algorithms, Harper Collins, 1997 9. Robert Sedgewick and Kevin Wayne, Algorithms, fourth edition, Addison Wesley, 2011. 10. Harsh Bhasin, "Algorithm Design and Analysis", First Edition, Oxford University Press. 11. Gilles Brassard and Paul Bratley, Algorithms: Theory and Practice, Prentice Hall, 1995.		

Evaluation Scheme

B.TECH.

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

CURRICULUM STRUCTURE

SEMESTER- V

SEMESTER- V													
Sl. No.	Subject Codes	Subject	Periods			Evaluation Scheme				End Semester		Total	Credit
			L	T	P	CT	TA	Total	PS	TE	PE		
1	BCS501	Database Management System	3	1	0	20	10	30		70		100	4
2	BCAI501	Artificial Intelligence	3	1	0	20	10	30		70		100	4
3	BCS503	Design and Analysis of Algorithm	3	1	0	20	10	30		70		100	4
4	BCAI051/ BCS058/ BCS052/ BCS054	Departmental Elective-I	3	1	0	20	10	30		70		100	4
5	BCAM051/ BCAI052/ BCS056/ BCS057	Departmental Elective-II	3	1	0	20	10	30		70		100	4
6	BCS551	Database Management System Lab	0	0	2				50		50	100	1
7	BCAI551	Artificial Intelligence Lab	0	0	2				50		50	100	1
8	BCS553	Design and Analysis of Algorithm Lab	0	0	2				50		50	100	1
9	BCS554	Mini Project or Internship Assessment*	0	0	2				50		50	100	2
10	BNC501/ BNC502	Constitution of India/ Essence of Indian Traditional Knowledge	2	0	0	20	10	30		70			
		Total										900	23

*The Mini Project or Internship (4 weeks) conducted during summer break after IV semester and will be assessed during V semester.

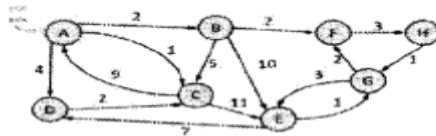
*It is desirable that the students should do their Summer Internship or Mini Project in their specialization area in line with the B.Tech. program.

Question Bank

Unit-III

Syllabus: Divide and Conquer with Examples Such as Sorting, Matrix Multiplication, Convex Hull and Searching. Greedy Methods with Examples Such as Optimal Reliability Allocation, Knapsack, Minimum Spanning Trees – Prim’s and Kruskal’s Algorithms, Single Source Shortest Paths- Dijkstra’s and Bellman Ford Algorithms.

1. What do you mean by minimum spanning “tree? Write an algorithm for minimum spanning tree that may generate multiple forest trees and also explain with suitable example.
2. Describe in detail the Strassen’s Matrix Multiplication algorithms based on divide and conquer strategies with suitable example.
3. Prove that if the weights on the edge of the connected undirected graph are distinct then there is a unique Minimum Spanning Tree. Give an example in this regard. Also discuss Prim’s Minimum Spanning Tree Algorithm in detail.
4. When do Dijkstra and the Bellman-Ford algorithm both fail to find a shortest path? Can Bellman ford detect all negative weight cycles in a graph? Apply Bellman Ford Algorithm on the following graph:



5. List out the disadvantages of divide and conquer algorithm.
6. Briefly explain the Prim’s algorithm.
7. Consider following instance for simple knapsack problem. Find the solution using greedy method. $N=8$ $P= \{11, 21, 31, 33, 43, 53, 55, 65\}$ $W: \{1, 11, 21, 23, 33, 43, 45, 55\}$ $M: 110$.
8. Find the shortest path in the below graph from the source vertex 1 to all other vertices by using Dijkstra’s algorithm.
9. What is Minimum Cost Spanning Tree? Explain Kruskal’s Algorithm and Find MST of the Graph. Also write its Time-Complexity
10. Define spanning tree. Write Kruskal’s algorithm for finding minimum cost spanning tree. Describe how Kruskal s algorithm is different from Prim s algorithm for finding minimum cost spanning free.
11. What do you mean by convex hull? Describe an algorithm that solves the convex hull problem. Find the time complexity of the algorithm.
12. Given the six items in the table below and a Knapsack with Weight 100, what is the solution to the Knapsack problem in all concepts. I.e. explain greedy all approaches and find the optimal solution.

ITEM ID	WEIGHT	VALUE	VALUE/WEIGHT
A	100	40	.4
B	50	35	.7
C	40	20	.5
D	20	4	.2
E	10	10	1
F	10	6	.6

Unit-IV

Syllabus: Dynamic Programming with Examples Such as Knapsack. All Pair Shortest Paths – Warshal's and Floyd's Algorithms, Resource Allocation Problem. Backtracking, Branch and Bound with Examples Such as Travelling Salesman Problem, Graph Coloring, n-Queen Problem, Hamiltonian Cycles and Sum of Subsets.

1. Differentiate between Dynamic programming and Greedy approach. What is 0/1 knapsack problem? Solve the following instance using Dynamic programming write the algorithm also. Knapsack Capacity=10 $P=\langle 1,6,18,22,28 \rangle$ and $w=\langle 1,2,5,6,7 \rangle$.

2. Differentiate between Backtracking and Branch and Bound approach. Write an algorithm for sum subset problem using back tracking approach. Find all possible solution for following instance using same if $m=30$, $s=\langle 1,2,5,7,8,10,15,20,25 \rangle$.

3. Define TSP problem in detail. Find the solution for the following instance of TSP problem using branch and bound.

4. Define principal of optimality, When and how dynamic programming is applicable. 5. Explain application of graph coloring problem. 7

6. Differences between branch & bound and backtracking technique.

7. Compare adjacency matrix and linked Adjacency lists representation of a Graph with suitable example/diagram.

8. Solve the Subset sum problem using Backtracking where $n=4$ $m=18$ $w[4] = \{5, 10, 8, 13\}$.

9. Give Floyd War shall algorithm to find the shortest path for all pairs of vertices in a graph. Give the complexity of the algorithm Explain with example

10. What is travelling salesman problem? Find the solution 'of following travelling salesman problem using branch and bound method. Cost matrix =

=

∞	20	30	10	11
15	∞	16	4	2
3	5	∞	2	4
19	6	18	∞	3
16	4	7	16	∞

11. Solve the following 0/1 knapsack problem using dynamic programming. $P=[11,21,31,33]$ $w=[2,11,22,15]$ $c=40$, $n=4$. 12. Find an LCS for the sequences. $X=\{x_1, x_2, \dots, x_m\}$ and $Y=\{y_1, y_2, \dots, y_n\}$. Also show that it requires $O(m+n)$ time.

Resource used

1. YouTube Videos:

SG Academy

<https://www.youtube.com/channel/UCXmoAxvVyJIR2dukvJQn1tQ>

2. Links of NPTEL lectures

<https://nptel.ac.in/courses/106/105/106105214/>

<https://nptel.ac.in/courses/106/102/106102132/>

<https://nptel.ac.in/courses/106/106/106106144/>

<https://nptel.ac.in/courses/106/108/106108101/>

3. Virtual Labs

<http://ebootathon.com/labs/beta/csit/OS/exp2/>